
Linux Environment

An **operating system** (OS) is the sum of all programs which are required to operate a computer and which control and monitor the application programs. **UNIX** was originally written in the programming language C, and it is mainly used for scientific-technical applications on mainframes and workstations but has become also perfectly suited for application in networks. Originally, it was a command-line oriented but can be used via a graphical user interface.

Linux is a family of UNIX-like open-source operating system developed by Linus Torvalds. Being an open-source operating system, there are instances that the source code of the Linux Kernel is used to form another operating system. **Linux distribution (distro)** is a collection of software on top of a Linux kernel. It can bundle server software, system management tools, documentation and many desktop applications in a central secure software repository. A distro aims to provide a common look and feel, secure and easy software management and often for a specific operational purpose.

Linux Environment

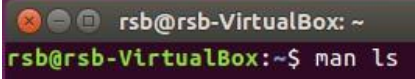
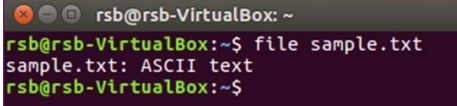
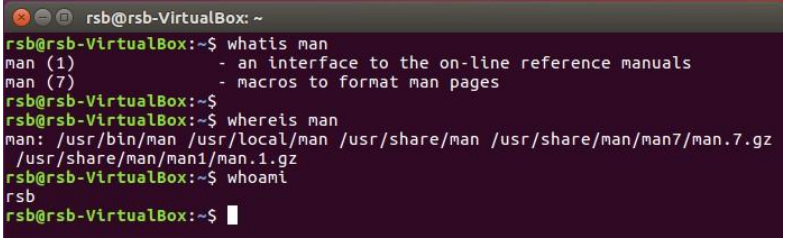
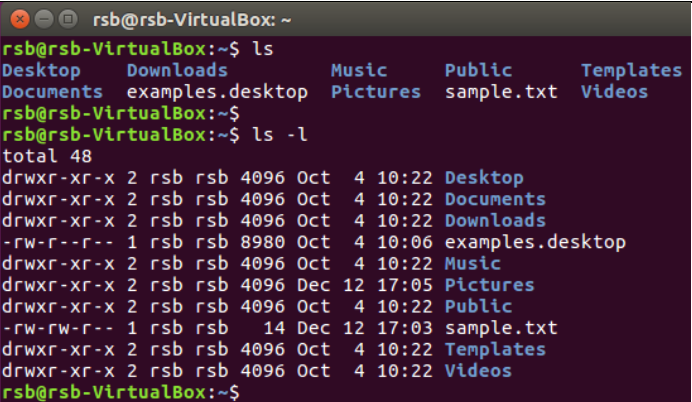
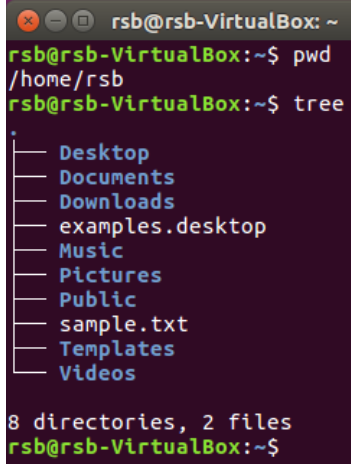
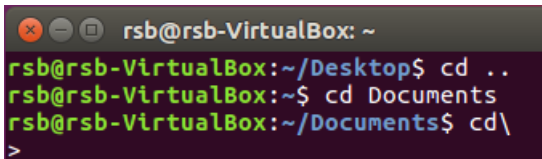
Shells are interfaces to the operating system the kernel of the host. It provides the necessary instruction and operation that an operating system may process. It can be in two forms:

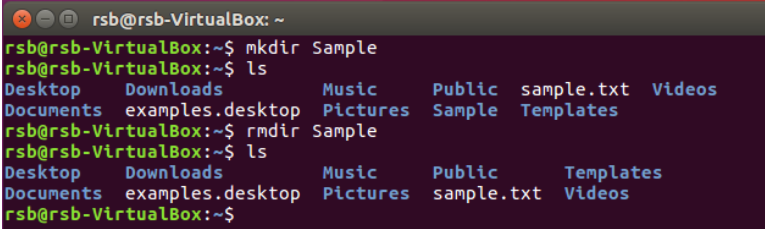
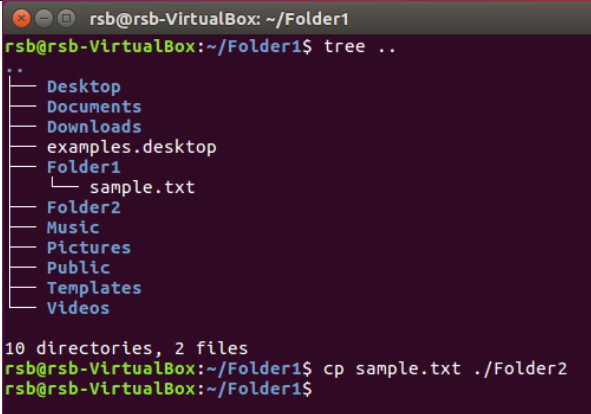
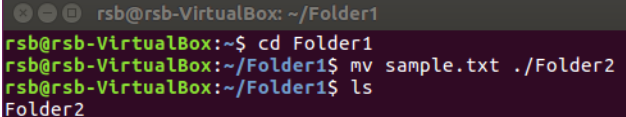
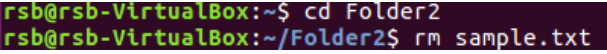
- A **GUI (Graphic User Interface)** is a graphical representation in which the users can interact with software or devices through graphical icons.
- A **CLI (Command Line Interface)** is a console or text-based representation in which the user types the commands to operate the software or devices.
- When it comes to ease of use, the new users will pick up a GUI much faster than a CLI. New users of CLI will have some difficulty operating it because they are not familiar with the commands.
- When it comes to controlling the system, CLI users will have all the control over the file system and operating system, and the tasks become simple through **scripting** while GUI users can do only simple and limited tasks not flexible enough for controlling the system.

Linux Desktop Environment (GUI)

- **GNOME** was once the most popular Linux desktop environment and was used by default on Ubuntu, Fedora, Debian, and most other big Linux distributions. It was a simple, fairly lightweight desktop environment.
- **KDE** has always been more complex than GNOME, packing in many more configuration options and features. It's a bit more Windows-like than the other desktop environments here, coming with a single taskbar on the bottom of the screen that includes a menu, quick launch-type icons, a taskbar, a notification area, and a clock — the typical layout of a Windows taskbar.
- **Xfce** is a more lightweight desktop environment, very similar to GNOME. This option is ideal if you want a more traditional desktop environment without full-screen application launchers, overdone graphical effects, and desktop widgets. It's also more lightweight than the other options here, making it ideal for older computers or ones without stable 3D graphics drivers that can't handle the effects in Unity and GNOME.
- **Cinnamon** was developed for Linux Mint in which it is based on GNOME. It uses up-to-date libraries and other software — but it takes that software and tries to create a more traditional-looking desktop with it. This modern desktop environment offers nice graphical effects and a rethought application menu.

Basic Linux Commands with Syntaxes

Commands	Description	Example Output:
man (Manual)	displays user manual of any command on the terminal	
file (Filetype)	determines the type of a file.	
whatis	displays short descriptions of program	
whereis	locates files and pages of a command	
whoami	displays the current user	
ls (List)	displays the list of files and directory 1st col. - permissions 2nd col. - no. of links 3rd col. - file/dir. owner 4th col. - file/dir. group 5th col. - size 6th col. - date stamp 7th col. - timestamp 8th col. - name	
tree	displays the contents of the working directory recursively showing sub-directories and files, and a summary of the total number of sub-directories and files	
pwd (Parent Working Directory)	displays parent working directory	
cd (Change Directory)	changes the current directory - cd .. - previous folder - cd / - root folder - cd ~ - shortcut	

mkdir (Make Directory)	creates directory	 <pre> rsb@rsb-VirtualBox: ~ rsb@rsb-VirtualBox:~\$ mkdir Sample rsb@rsb-VirtualBox:~\$ ls Desktop Downloads Music Public sample.txt Videos Documents examples.desktop Pictures Sample Templates rsb@rsb-VirtualBox:~\$ rmdir Sample rsb@rsb-VirtualBox:~\$ ls Desktop Downloads Music Public Templates Documents examples.desktop Pictures sample.txt Videos rsb@rsb-VirtualBox:~\$ </pre>
rmdir (Remove Directory)	removes directory	
cp (Copy File)	copies a file to a specified location	 <pre> rsb@rsb-VirtualBox: ~/Folder1 rsb@rsb-VirtualBox:~/Folder1\$ tree ├── Desktop ├── Documents ├── Downloads ├── examples.desktop ├── Folder1 │ └── sample.txt ├── Folder2 ├── Music ├── Pictures ├── Public ├── Templates └── Videos 10 directories, 2 files rsb@rsb-VirtualBox:~/Folder1\$ cp sample.txt ./Folder2 rsb@rsb-VirtualBox:~/Folder1\$ </pre>
mv (Move File)	moves a file to a specified location	 <pre> rsb@rsb-VirtualBox: ~/Folder1 rsb@rsb-VirtualBox:~\$ cd Folder1 rsb@rsb-VirtualBox:~/Folder1\$ mv sample.txt ./Folder2 rsb@rsb-VirtualBox:~/Folder1\$ ls Folder2 </pre>
rm (Remove File)	removes a file.	 <pre> rsb@rsb-VirtualBox:~\$ cd Folder2 rsb@rsb-VirtualBox:~/Folder2\$ rm sample.txt </pre>

File Management and Structure

All files are organized into directories. These directories are organized into a tree-like structure called the filesystem. In Linux, there are three basic types of files:

- **Ordinary Files** – An ordinary file is a file on the system that contains data, text, or program instructions. In this tutorial, you look at working with ordinary files.
- **Directories** – Directories store both special and ordinary files. For users familiar with Windows or Mac OS, Unix directories are equivalent to folders.
- **Special Files** – Some special files provide access to hardware such as hard drives, CD-ROM drives, modems, and Ethernet adapters. Other special files are similar to aliases or shortcuts and enable you to access a single file using different names.

Linux Directories

- **/** (root directory) refers to the *root* directory. The root directory is the one from which all other directories branch off from.
- **/bin** is the directory that contains *binaries*, that is, some of the applications and programs you can run.
- **/boot** directory contains files required for starting your system.
- **/dev** contains *device* files. Many of these are generated at boot time or even on the fly.
- **/etc** is the dumping ground for system files administrators were not sure where else to put.
- **/home** is where you will find your users' personal directories.
- **/lib** is where *libraries* live. Libraries are files containing code that your applications can use.

They contain snippets of code that applications use to draw windows on your desktop, control peripherals, or send files to your hard disk.

- **/opt** directory is often where the software you compile. Applications will end up in the **/opt/bin** directory and libraries in the **/opt/lib** directory.
- **/proc** and **/dev** are virtual directory in which it contains information about your computers, such as information about your CPU and the kernel your Linux system is running. It is a directory in which it contains the files and directories are generated when your computer starts, or on the fly, as the system is running and things change.
- **/sbin** is similar to **/bin**, but it contains applications that only the superuser will need. You can use these applications with the **sudo**

command that temporarily concedes you superuser powers on many distributions.

- **/usr** directory was where users' home directories were originally kept back in the early days of UNIX. It contains a mish-mash of directories which in turn contain applications, libraries, documentation, wallpapers, icons and a long list of other stuff that needs to be shared by applications and services.
- **/tmp** contains temporary files, usually placed there by applications that you are running. The files and directories often (not always) contain data that an application doesn't need right now, but may need later on.
- **/var** was originally given its name because its contents were deemed *variable* in which it contains files to which the system writes data during the course of its operation.

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